CHAPTER A

Supervised tests

Two versions are provided for each chapter. The reference structure is: fluency questions worth 6 points, an exercise worth 6 points and a problem worth 8 points.

Chapter 1 – Numerical sequences

Supervised test 1 · Version A

Numerical sequences **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, communicate

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.1 · 1 – Fluency. [6 points]

1. Give the multiplier associated with an increase of 5%.
2. Is the sequence $12, 17, 22, 27, \dots$ arithmetic? Give its common difference.
3. How many terms does $u_4 + u_5 + \cdots + u_{18}$ contain?
4. If $w_0 = 80$ and $w_{n+1} = 0.9 w_n$, calculate w_2.
5. Write $1 + 2 + \cdots + 20$ using the symbol $\sum$.

Exercise A.2 · 2 – Subscriptions. [6 points] A platform has 120,000 subscribers at the start of the year and gains 11,000 each month. Let u_n be the number of subscribers, in thousands, after n months.

1. Show that (u_n) is arithmetic and state its common difference.
2. Express u_n in terms of n, then calculate u_{12}.
3. Calculate $u_1 + \cdots + u_{12}$. Interpret this sum with caution.

Exercise A.3 · 3 – Investment and target. [8 points] An initial capital of 650,000 Ar increases by 5% per year. Let C_n be the capital after n years.

1. Write C_{n+1} in terms of C_n, then C_n in terms of n.
2. Calculate C_5, rounded to the nearest ariary.
3. Use a calculator to find the first integer n for which C_n exceeds 1,000,000 Ar.
4. Explain why the model is geometric and not arithmetic.

Solution. **Marking scheme.** Ex. 1: 6 points spread over the fluency questions. Ex. 2: 6 points. Ex. 3: 8 points.
Solution outline. The multiplier is 1.05. The first sequence in the fluency questions is arithmetic with common difference 5; the sum contains 15 terms; $w_2 = 64$ and $\sum_{k=1}^{20} k$. For exercise 2, $u_n = 120 + 11n$, so $u_{12} = 252$; the requested sum uses $12\dfrac{u_1 + u_{12}}{2}$ and accumulates monthly figures, not a number of distinct people. For exercise 3, $C_n = 650{,}000 \times \left(1 + \dfrac{5}{100}\right)^n$; check the threshold with two consecutive values.

Supervised test 1 · Version B

Numerical sequences **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, communicate

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.4 · 1 – Fluency. [6 points]

1. Give the multiplier associated with an increase of 4%.
2. Is the sequence $12, 17, 22, 27, \ldots$ arithmetic? Give its common difference.
3. How many terms does $u_4 + u_5 + \cdots + u_{18}$ contain?
4. If $w_0 = 80$ and $w_{n+1} = 0.9w_n$, calculate w_2.
5. Write $1 + 2 + \cdots + 20$ using the symbol $\sum$.

Exercise A.5 · 2 – Subscriptions. [6 points] A platform has 145,000 subscribers at the start of the year and gains 9,000 each month. Let u_n be the number of subscribers, in thousands, after n months.

1. Show that (u_n) is arithmetic and state its common difference.
2. Express u_n in terms of n, then calculate u_{12}.
3. Calculate $u_1 + \cdots + u_{12}$. Interpret this sum with caution.

Exercise A.6 · 3 – Investment and target. [8 points] An initial capital of 720,000 Ar increases by 4% per year. Let C_n be the capital after n years.

1. Write C_{n+1} in terms of C_n, then C_n in terms of n.
2. Calculate C_5, rounded to the nearest ariary.
3. Use a calculator to find the first integer n for which C_n exceeds 1,000,000 Ar.
4. Explain why the model is geometric and not arithmetic.

Solution. **Marking scheme.** Ex. 1: 6 points spread over the fluency questions. Ex. 2: 6 points. Ex. 3: 8 points.
Solution outline. The multiplier is 1.04. The first sequence in the fluency questions is arithmetic with common difference 5; the sum contains 15 terms; $w_2 = 64$ and $\sum_{k=1}^{20} k$. For exercise 2, $u_n = 145 + 9n$, so $u_{12} = 253$; the requested sum uses $12\dfrac{u_1 + u_{12}}{2}$ and accumulates monthly figures, not a number of distinct people. For exercise 3, $C_n = 720{,}000 \times \left(1 + \dfrac{4}{100}\right)^n$; check the threshold with two consecutive values.

Chapter 2 – Exponential functions

Supervised test 2 · Version A

Exponential functions **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: calculate, represent, model, interpret

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.7 · 1 – Powers and variations. [6 points]

1. Simplify $2^4 \times 2^{-7}$ and $\dfrac{5^{x+3}}{5^x}$.
2. Write 3^{-2} as a fraction.
3. Give the direction of variation of $f(x) = 1.04^x$ and of $g(x) = -2 \times 0.75^x$.
4. Explain why $a^x + a^y = a^{x+y}$ is false in general.

Exercise A.8 · 2 – Website traffic. [6 points] At launch, a website receives 2400 visits per day. Traffic increases by 4% per week. It is modelled by $V(t) = 2400 \times 1.04^t$.

1. Interpret the numbers 2400 and 1.04.
2. Calculate $V(8)$, rounded to the nearest unit.
3. Find the first whole number of weeks for which $V(t) > 3100$.

Exercise A.9 · 3 – Average rate. [8 points] A company's turnover goes from 82,000 to 104000 euros in four years.

1. Calculate the overall multiplier.
2. Write the average annual multiplier as a power.
3. Calculate the average annual rate of change (percentage change), rounded to 0.01%.
4. Explain why the overall rate is not simply divided by four.

Solution. **Marking scheme:** $6 + 6 + 8$. **Solution:** $2^4 \times 2^{-7} = 2^{-3} = \dfrac{1}{8}$, $\dfrac{5^{x+3}}{5^x} = 125$, $3^{-2} = \dfrac{1}{9}$. f is increasing and g is increasing. In the problem, calculate $V(8)$ then test two consecutive integers around the threshold. For the average rate, $q = \left(\dfrac{V_f}{V_i}\right)^{\frac{1}{4}}$ and $t = q - 1$; multipliers multiply.

Supervised test 2 · Version B

Exponential functions **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: calculate, represent, model, interpret

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.10 · 1 – Powers and variations. [6 points]

1. Simplify $2^4 \times 2^{-7}$ and $\dfrac{5^{x+3}}{5^x}$.
2. Write 3^{-2} as a fraction.
3. Give the direction of variation of $f(x) = 1.03^x$ and of $g(x) = -2 \times 0.75^x$.
4. Explain why $a^x + a^y = a^{x+y}$ is false in general.

Exercise A.11 · 2 – Website traffic. [6 points] At launch, a website receives 1850 visits per day. Traffic increases by 3% per week. It is modelled by $V(t) = 1850 \times 1.03^t$.

1. Interpret the numbers 1850 and 1.03.
2. Calculate $V(8)$, rounded to the nearest unit.
3. Find the first whole number of weeks for which $V(t) > 2250$.

Exercise A.12 · 3 – Average rate. [8 points] A company's turnover goes from 82,000 to 98000 euros in four years.

1. Calculate the overall multiplier.
2. Write the average annual multiplier as a power.
3. Calculate the average annual rate of change (percentage change), rounded to 0.01%.
4. Explain why the overall rate is not simply divided by four.

Solution. **Marking scheme:** $6 + 6 + 8$. **Solution:** $2^4 \times 2^{-7} = 2^{-3} = \dfrac{1}{8}$, $\dfrac{5^{x+3}}{5^x} = 125$, $3^{-2} = \dfrac{1}{9}$. f is increasing and g is increasing. In the problem, calculate $V(8)$ then test two consecutive integers around the threshold. For the average rate, $q = \left(\dfrac{V_f}{V_i}\right)^{\frac{1}{4}}$ and $t = q - 1$; multipliers multiply.

Chapter 3 – The common logarithm function

Supervised test 3 · Version A

Common logarithm **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: calculate, solve, reason, communicate

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.13 · 1 – Knowledge and calculations. [6 points]

1. Calculate $\log(1000)$, $\log(0.01)$ and $\log(10^{-4})$.
2. For $x, y > 0$, expand $\log\left(\dfrac{100x^3}{y}\right)$.
3. Solve $10^t = 28000$ and give t to the nearest hundredth.

Exercise A.14 · 2 – Growth target. [6 points] A value is multiplied by 1.05 each year. We look for the time needed for it to double.

1. Translate the problem into an equation.
2. Solve this equation using the common logarithm (base 10).
3. Give the first whole number of years that reaches or exceeds double.
4. Check with the two relevant consecutive powers.

Exercise A.15 · 3 – Sound level. [8 points] The sound level is $L = 10\log\left(\dfrac{I}{I_0}\right)$, in decibels.

1. Calculate L when $I = 10^6 I_0$.
2. For $L = 84$, express then calculate the ratio $\dfrac{I}{I_0}$.
3. Compare the intensities corresponding to 84 dB and 74 dB.

Solution. **Marking scheme:** $6 + 6 + 8$. **Solution:** The usual values are $3, -2, -4$. The expansion gives $2 + 3\log x - \log y$. $t = \log(28000)$. For the doubling, $x = \dfrac{\log 2}{\log(1.05)}$ then take the next integer up and check. $\dfrac{I}{I_0} = 10^{\frac{L}{10}}$; a difference of 10 dB corresponds to an intensity multiplied by 10.

Supervised test 3 · Version B

Common logarithm **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: calculate, solve, reason, communicate

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.16 · 1 – Knowledge and calculations. [6 points]

1. Calculate $\log(1000)$, $\log(0.01)$ and $\log(10^{-4})$.
2. For $x, y > 0$, expand $\log\left(\dfrac{100x^3}{y}\right)$.
3. Solve $10^t = 32000$ and give t to the nearest hundredth.

Exercise A.17 · 2 – Growth target. [6 points] A value is multiplied by 1.07 each year. We look for the time needed for it to double.

1. Translate the problem into an equation.
2. Solve this equation using the common logarithm (base 10).
3. Give the first whole number of years that reaches or exceeds double.
4. Check with the two relevant consecutive powers.

Exercise A.18 · 3 – Sound level. [8 points] The sound level is $L = 10\log\left(\dfrac{I}{I_0}\right)$, in decibels.

1. Calculate L when $I = 10^6 I_0$.
2. For $L = 78$, express then calculate the ratio $\dfrac{I}{I_0}$.
3. Compare the intensities corresponding to 78 dB and 68 dB.

Solution. **Marking scheme:** $6 + 6 + 8$. **Solution:** The usual values are $3, -2, -4$. The expansion gives $2 + 3\log x - \log y$. $t = \log(32000)$. For the doubling, $x = \dfrac{\log 2}{\log(1.07)}$ then take the next integer up and check. $\dfrac{I}{I_0} = 10^{\frac{L}{10}}$; a difference of 10 dB corresponds to an intensity multiplied by 10.

Chapter 4 – The reciprocal function

Supervised test 4 · Version A

Reciprocal function **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: calculate, reason, represent, communicate

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.19 · 1 – Reciprocal function. [6 points] Consider $f(x) = 2 + \dfrac{16}{x}$.

1. Give its domain.
2. Calculate its derivative and study its sign.
3. Give its asymptotes and its direction of variation on each interval of the domain.

Exercise A.20 · 2 – Sign table. [6 points] Solve $\dfrac{(x-4)(x+2)}{x} \geq 0$. The table must show the zeros and the excluded value.

Exercise A.21 · 3 – Average cost. [8 points] For $x > 0$, an average cost is modelled by $C(x) = x + \dfrac{16}{x} + 12$.

1. Calculate $C'(x)$ and factorise its numerator.
2. Draw up the variation table of C on $]0, +\infty[$.
3. Find the value of x that minimises the average cost and calculate this minimum.
4. Sketch a consistent graph of the curve.

Solution. **Marking scheme:** $6 + 6 + 8$. **Solution:** $D_f = \mathbb{R} \setminus \{0\}$, $f'(x) = -\dfrac{16}{x^2} < 0$, asymptotes $x = 0$ and $y = 2$. For the quotient, order $-2, 0, 4$ and exclude 0. For the cost, $C'(x) = 1 - \dfrac{16}{x^2} = \dfrac{(x-4)(x+4)}{x^2}$; minimum at $x = 4$, with value 20.

Supervised test 4 · Version B

Reciprocal function **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: calculate, reason, represent, communicate

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.22 · 1 – Reciprocal function. [6 points] Consider $f(x) = 2 + \dfrac{9}{x}$.

1. Give its domain.
2. Calculate its derivative and study its sign.
3. Give its asymptotes and its direction of variation on each interval of the domain.

Exercise A.23 · 2 – Sign table. [6 points] Solve $\dfrac{(x-3)(x+2)}{x} \geq 0$. The table must show the zeros and the excluded value.

Exercise A.24 · 3 – Average cost. [8 points] For $x > 0$, an average cost is modelled by $C(x) = x + \dfrac{9}{x} + 12$.

1. Calculate $C'(x)$ and factorise its numerator.
2. Draw up the variation table of C on $]0, +\infty[$.
3. Find the value of x that minimises the average cost and calculate this minimum.
4. Sketch a consistent graph of the curve.

Solution. **Marking scheme:** $6 + 6 + 8$. **Solution:** $D_f = \mathbb{R} \setminus \{0\}$, $f'(x) = -\dfrac{9}{x^2} < 0$, asymptotes $x = 0$ and $y = 2$. For the quotient, order $-2, 0, 3$ and exclude 0. For the cost, $C'(x) = 1 - \dfrac{9}{x^2} = \dfrac{(x-3)(x+3)}{x^2}$; minimum at $x = 3$, with value 18.

Chapter 5 – Statistics with two quantitative variables

Supervised test 5 · Version A

Two-variable statistics **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: represent, model, calculate, interpret

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.25 · 1 – Constructing and describing. [6 points] We observe the pairs $(1,8)$, $(2,11)$, $(3,13)$, $(4,17)$, $(5,19)$ and $(6,23)$.

1. Draw the scatter plot on a correctly labelled coordinate system.
2. Calculate the coordinates of the mean point.
3. Describe the trend and the spread without concluding that there is causality.

Exercise A.26 · 2 – Using a linear regression. [6 points] A linear regression line is $y = 3.2x + 12$, established for $x \in [2, 15]$.

1. Estimate y for $x = 10$ and characterise the estimate.
2. Estimate x when $y = 60$.
3. Estimate y for $x = 40$, then explain why this result must be treated with caution.

Exercise A.27 · 3 – Change of variable. [8 points] Let $z = \log(y)$, and the linear regression of (x, z) is $z = 0.06x + 1.1$.

1. Express y in terms of x.
2. Estimate y for $x = 8$.
3. Explain the benefit of the change of variable and one limitation of extrapolation.

Solution. **Marking scheme:** $6 + 6 + 8$. **Solution:** Calculate $\bar{x}$ and $\bar{y}$ from the six values. With the line, substitute x or solve an equation; $x = 10$ is an interpolation and $x = 40$ a distant extrapolation. The final model is $y = 10^{0.06x+1.1}$.

Supervised test 5 · Version B

Two-variable statistics **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: represent, model, calculate, interpret

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.28 · 1 – Constructing and describing. [6 points] We observe the pairs $(1,8)$, $(2,11)$, $(3,13)$, $(4,17)$, $(5,22)$ and $(6,23)$.

1. Draw the scatter plot on a correctly labelled coordinate system.
2. Calculate the coordinates of the mean point.
3. Describe the trend and the spread without concluding that there is causality.

Exercise A.29 · 2 – Using a linear regression. [6 points] A linear regression line is $y = 2.4x + 18$, established for $x \in [2, 15]$.

1. Estimate y for $x = 10$ and characterise the estimate.
2. Estimate x when $y = 60$.
3. Estimate y for $x = 40$, then explain why this result must be treated with caution.

Exercise A.30 · 3 – Change of variable. [8 points] Let $z = \log(y)$, and the linear regression of (x, z) is $z = 0.06x + 1.1$.

1. Express y in terms of x.
2. Estimate y for $x = 8$.
3. Explain the benefit of the change of variable and one limitation of extrapolation.

Solution. **Marking scheme:** $6 + 6 + 8$. **Solution:** Calculate $\bar{x}$ and $\bar{y}$ from the six values. With the line, substitute x or solve an equation; $x = 10$ is an interpolation and $x = 40$ a distant extrapolation. The final model is $y = 10^{0.06x+1.1}$.

Chapter 6 – Conditional probability

Supervised test 6 · Version A

Conditional probability **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, interpret

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.31 · 1 – Weighted tree. [6 points] A proportion 0.35 of orders are express. The late-delivery rate is 0.05 among express orders and 0.12 among the others.

1. Define the events E and R, then build the complete weighted tree.
2. Calculate $P(E \cap R)$.
3. Calculate $P(R)$ using the law of total probability.
4. Calculate $P_R(E)$ and interpret.

Exercise A.32 · 2 – Independence. [6 points] In another study, $P(A) = 0.4$, $P(B) = 0.25$ and $P(A \cap B) = 0.10$.

1. Calculate $P(A)P(B)$.
2. Are the events independent? Justify.
3. Calculate $P_A(B)$ and confirm the conclusion.

Exercise A.33 · 3 – Frequency table. [8 points] Out of 800 customers, 320 have a premium card; 96 premium customers and 72 non-premium customers cancelled. Build the complete table, then calculate the conditional probability of being premium given that the customer cancelled.

Solution. **Marking scheme:** $8 + 6 + 6$. **Solution:** $P(R) = P(E)P_E(R) + P(\overline{E})P_{\overline{E}}(R)$ then $P_R(E) = \dfrac{P(E \cap R)}{P(R)}$. For independence, compare the intersection with the product 0.10. In the table, there are 168 cancellations, so $P_R(P) = \dfrac{96}{168} = \dfrac{4}{7}$.

Supervised test 6 · Version B

Conditional probability **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, interpret

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.34 · 1 – Weighted tree. [6 points] A proportion 0.45 of orders are express. The late-delivery rate is 0.08 among express orders and 0.15 among the others.

1. Define the events E and R, then build the complete weighted tree.
2. Calculate $P(E \cap R)$.
3. Calculate $P(R)$ using the law of total probability.
4. Calculate $P_R(E)$ and interpret.

Exercise A.35 · 2 – Independence. [6 points] In another study, $P(A) = 0.4$, $P(B) = 0.25$ and $P(A \cap B) = 0.14$.

1. Calculate $P(A)P(B)$.
2. Are the events independent? Justify.
3. Calculate $P_A(B)$ and confirm the conclusion.

Exercise A.36 · 3 – Frequency table. [8 points] Out of 800 customers, 320 have a premium card; 96 premium customers and 72 non-premium customers cancelled. Build the complete table, then calculate the conditional probability of being premium given that the customer cancelled.

Solution. **Marking scheme:** $8 + 6 + 6$. **Solution:** $P(R) = P(E)P_E(R) + P(\overline{E})P_{\overline{E}}(R)$ then $P_R(E) = \dfrac{P(E \cap R)}{P(R)}$. For independence, compare the intersection with the product 0.10. In the table, there are 168 cancellations, so $P_R(P) = \dfrac{96}{168} = \dfrac{4}{7}$.

Chapter 7 – Discrete random variables and the binomial distribution

Supervised test 7 · Version A

Random variables and the binomial distribution **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, interpret

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.37 · 1 – Expected value of a game. [6 points] A game costs 3 euros. The gross winnings are 0, 5 or 15 euros with probabilities 0.65, 0.25 and 0.10.

1. Build the distribution of the net gain G.
2. Calculate $E(G)$.
3. Is the game favourable to the player? Interpret without claiming what will happen in a single play.

Exercise A.38 · 2 – Binomial distribution. [6 points] A customer responds favourably with probability 0.30. An adviser contacts 10 customers independently. Let X be the number of favourable responses.

1. Justify the distribution of X and give its parameters.
2. Calculate $P(X = 2)$.
3. Calculate $P(X = 0)$ then $P(X \geq 1)$.
4. Calculate and interpret $E(X)$.

Exercise A.39 · 3 – Pascal's triangle. [8 points]

1. Write row $n = 6$ of Pascal's triangle.
2. Read off $\binom{6}{2}$ and $\binom{6}{4}$.
3. Explain the role of $\binom{n}{k}$ in the binomial formula.

Solution. **Marking scheme:** $6 + 8 + 6$. **Solution:** The net gain is $-3, 2, 12$; $E(G) = -1.95 + 0.50 + 1.20 = -0.25$ euro. $X \sim \mathcal{B}(10, 0.30)$; apply the formula, use $1 - P(X = 0)$ for "at least one", and $E(X) = np$. Row $n = 6$ is $1, 6, 15, 20, 15, 6, 1$.

Supervised test 7 · Version B

Random variables and the binomial distribution **Duration: 1 h** **Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, interpret

A calculator is allowed unless stated otherwise. Every step must be legible and results must be put in context.

Exercise A.40 · 1 – Expected value of a game. [6 points] A game costs 3 euros. The gross winnings are 0, 5 or 15 euros with probabilities 0.65, 0.25 and 0.10.

1. Build the distribution of the net gain G.
2. Calculate $E(G)$.
3. Is the game favourable to the player? Interpret without claiming what will happen in a single play.

Exercise A.41 · 2 – Binomial distribution. [6 points] A customer responds favourably with probability 0.25. An adviser contacts 8 customers independently. Let X be the number of favourable responses.

1. Justify the distribution of X and give its parameters.
2. Calculate $P(X = 2)$.
3. Calculate $P(X = 0)$ then $P(X \geq 1)$.
4. Calculate and interpret $E(X)$.

Exercise A.42 · 3 – Pascal's triangle. [8 points]

1. Write row $n = 6$ of Pascal's triangle.
2. Read off $\binom{6}{2}$ and $\binom{6}{4}$.
3. Explain the role of $\binom{n}{k}$ in the binomial formula.

Solution. **Marking scheme:** $6 + 8 + 6$. **Solution:** The net gain is $-3, 2, 12$; $E(G) = -1.95 + 0.50 + 1.20 = -0.25$ euro. $X \sim \mathcal{B}(8, 0.25)$; apply the formula, use $1 - P(X = 0)$ for "at least one", and $E(X) = np$. Row $n = 6$ is $1, 6, 15, 20, 15, 6, 1$.